Automatic and optimized firewall reconfiguration

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Network Configuration Updates

Updates in modern networks are (often) **frequent** and **small**.

- <1% affected forwarding rules and ACL rules [1]

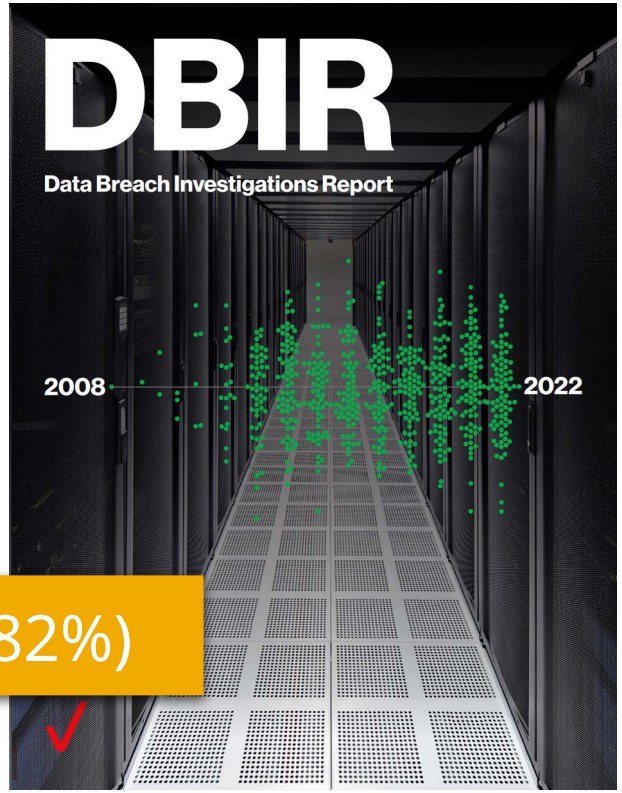
- 157 configuration lines on Facebook's network with a frequency of 12 changes/week (per device!)[2]

[1] D. Plonka and A. J. Tack, "*An Analysis of Network Configuration Artifacts*", Proc. of 23rd Large Installation System Administration Conference, USENIX Association, 2009
[2] Y.-W. E. Sung et al., "*Robotron: Top-down network management at Facebook scale*," in Proc. of the 2016 ACM SIGCOMM Conference, ser. SIGCOMM '16. New York, NY, 2016

Network Security Automation

Network Security Automation [3] is **necessary** to:

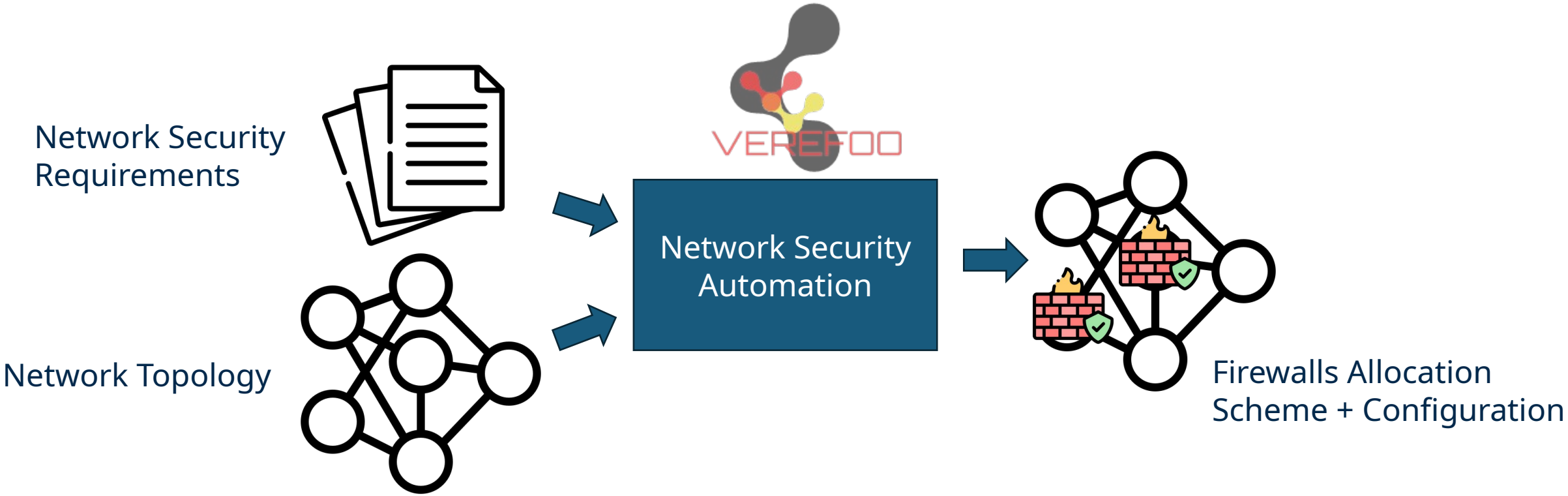
- Effectively manage complex networks
- Avoid trial-and-error and human involvement
- Additional features:
  - *Policy-based management*
  - *Formal verification*
  - *Optimization*



Human error (82%)

[3] D. Bringhenti, G. Marchetto, R. Sisto, and F. Valenza, "*Automation for network security configuration: State of the art and research trends*," ACM Comput. Surv., vol. 56, no. 3, pp. 57:1–57:37, 2024

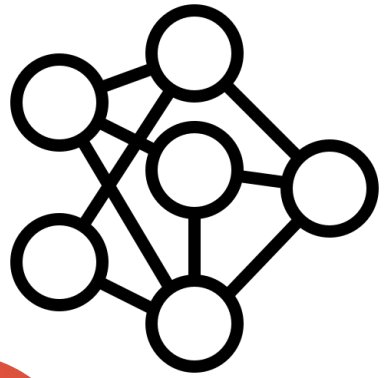
The Existing Approach



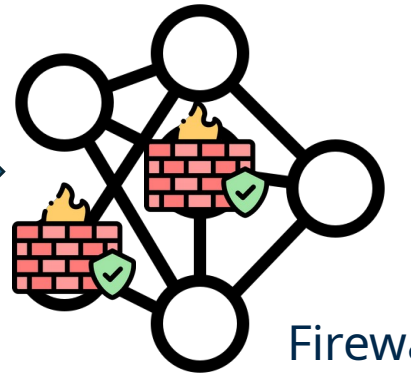
The Existing Approach

UPDATED
Network Security Requirements

Network Topology



Network Security Automation



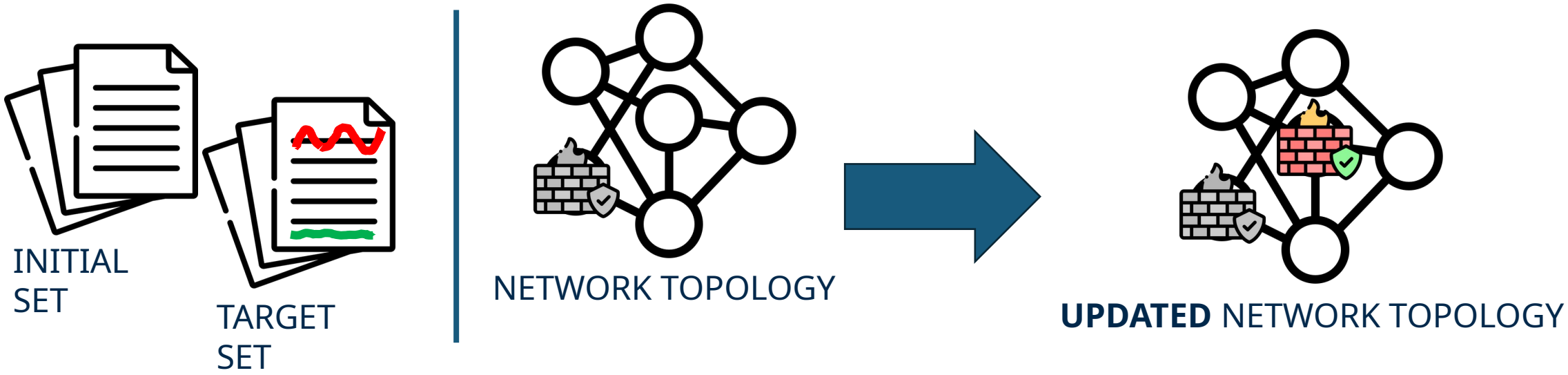
Firewalls Allocation Scheme + Configuration

When policies are updated (e.g., to mitigate an attack) the configuration of whole network must be recomputed **from scratch**

Contributions

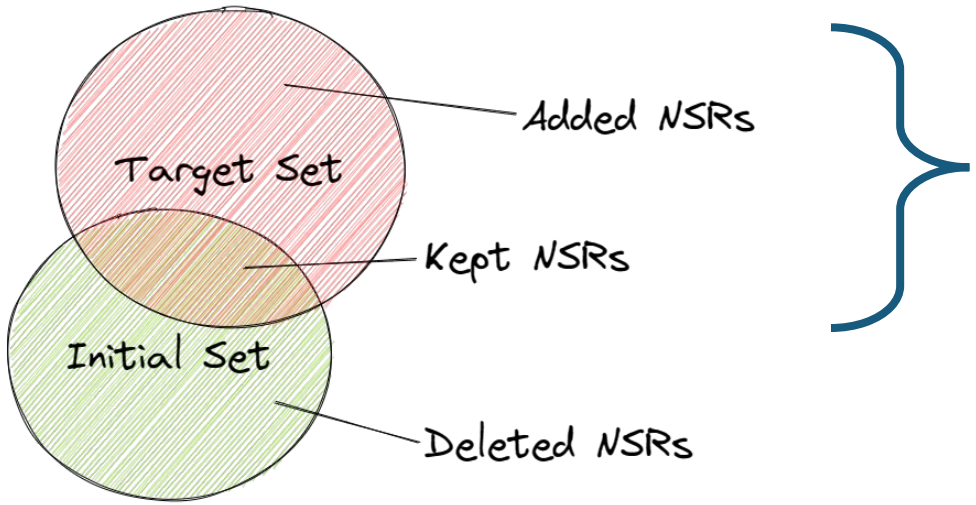
Automated approach for the *optimal* **reconfiguration** of distributed firewalls

- within a short computation time
- (formally) satisfying a **new set of Network Security Requirements** (NSRs)



Problem Inputs

- **Two sets of NSRs** + (partially configured) **Network Topology**



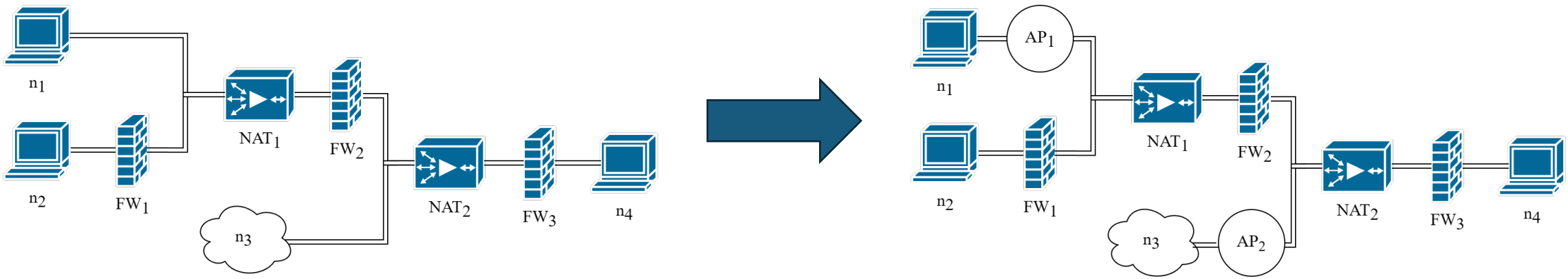
- Elements to reconfigure are selected starting from the *Added NSRs* (either new requirements or updated ones).
- Problem models *Added NSRs + Kepts NSRs* to ensure their satisfaction within the solution.

Type	IPSrc	IPDst	pSrc	pDst	tProto
Isol	192.168.0.2	220.226.50.∗	∗	∗	∗
Isol	192.168.0.3	220.226.50.∗	∗	∗	∗
Isol	130.192.225.∗	220.226.50.3	∗	∗	∗
Reach	130.192.225.∗	220.226.50.2	∗	80	TCP
Isol	130.192.225.∗	220.226.50.2	∗	≠80	TCP
Isol	130.192.225.∗	220.226.50.2	∗	∗	UDP
Reach	220.226.50.2	130.192.225.∗	∗	∗	∗
Isol	130.192.120.∗	220.226.50.2	∗	∗	∗
Reach	130.192.120.∗	220.226.50.3	∗	110	TCP
Isol	130.192.120.∗	220.226.50.3	∗	≠110	TCP
Isol	130.192.120.∗	220.226.50.3	∗	∗	UDP
Reach	220.226.50.3	130.192.120.∗	∗	∗	∗
Isol	130.192.120.∗	130.192.225.∗	∗	∗	∗

Each NSR r is modeled as a tuple (a, C), where a is the action to be applied on packets matching with the condition predicate C

Problem Inputs

- **Two sets of NSRs** + (partially configured) **Network Topology**
  - *Service Graph*, i.e., a directed graph composed of edges and vertices that represents the current network configuration.
  - This is automatically transformed into an *Allocation Graph* by placing an **Allocation Place** (AP) on every edge not including a FW.
    - APs represent the possible positions where a FW could be allocated.

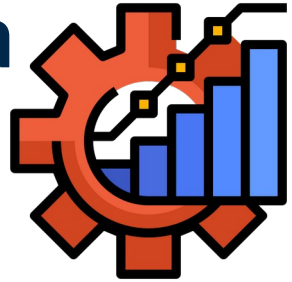


MaxSMT problem

A Maximum Satisfiability Modulo Theories (**MaxSMT**) problem differs from an SMT one because it allows the definition of two clauses:

- *Hard constraints* - compulsory
- *Soft constraints* – optional and with an associated weight

Optimization's criteria



NSRs

Hard Constraints

- Hard constraints are used to express NSRs.
- Working with packet filtering FWs, we model <u>connectivity requirements</u>:

Reachability

$$\exists f \in F_r . \forall i \left(n_i \in \pi(f) \wedge allocated(n_i) \Rightarrow \neg deny_i(\tau(f, n_i)) \right)$$

Isolation

$$\forall f \in F_r . \exists\, i . \left(n_i \in \pi(f) \wedge allocated(n_i) \wedge deny_i(\tau(f, n_i)) \right)$$

Algorithm

- Novel algorithm to select the **minimum** network area that must be reconfigured; other configuration components are kept as static elements.

- Implements a custom logic for each NSR type, i.e., Isolation or Reachability, to find all network elements to reconfigure because in conflict with at least a new NSR (i.e., *added*).

- As a result, we are **reducing the solution space** considered by the solver!

Soft Constraints

- Soft constraints are used for optimization. The two goals are:
  - *Resource minimization (i.e., number of FWs and FW rules)*
  - *Configuration state preservation*

- **Allocation** Soft Constraint

$$\forall k \mid a_k \in A_A . Soft(allocated(a_k) = false, c_k)$$

- **Configuration** Soft Constraint

$$\forall i \mid p_i \in P_k . Soft(configured(p_i) = false, c_{ki})$$

For each reconfigured FW, the previous have different weights to favor their allocation/configuration.

$$c_k^R, c_{ki}^R \mid (c_k^R < c_k) \wedge (c_{ki}^R < c_{ki})$$

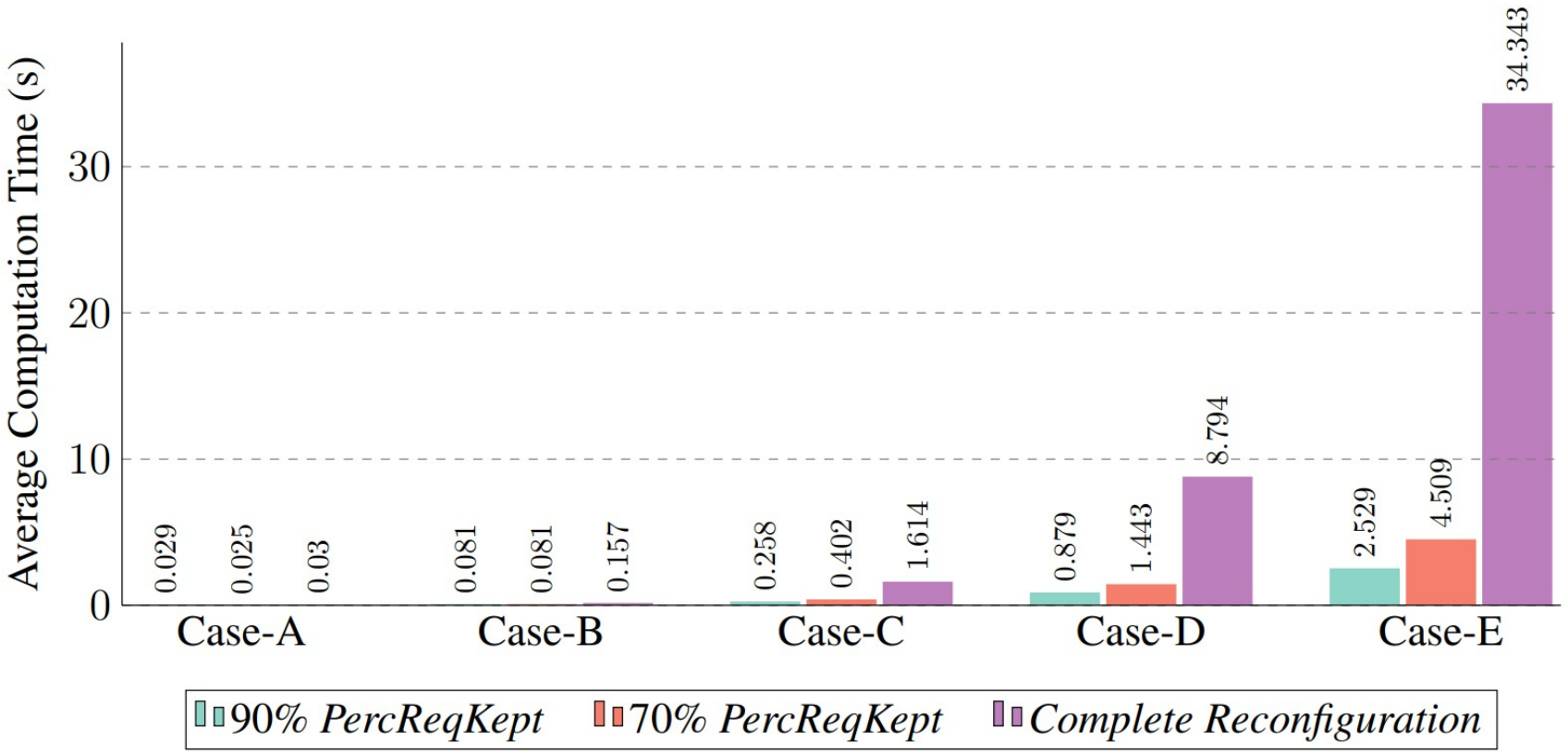
Implementation

Implementation:

- Java-based framework, with REST APIs
- z3 as MaxSMT solver

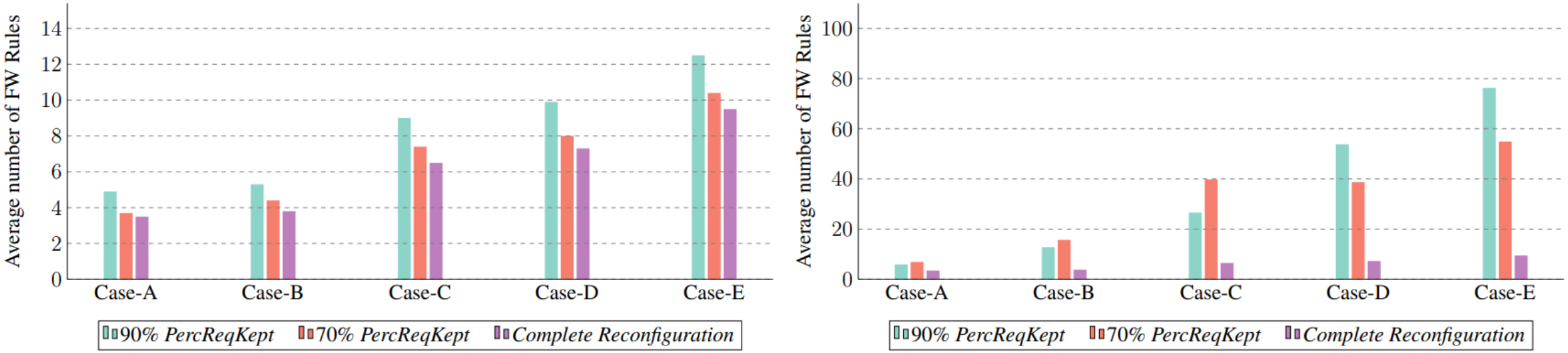
Code publicly available on GitHub (https://github.com/netgroup-polito/verefoo/tree/Dublin)

Validation - Performance



case-A: 10 NSRs, 60 endpoints, 5 NATs
case-B: 15 NSRs, 80 endpoints,10 NATs
case-C: 20 NSRs, 100 endpoints, 15 NATs
case-D: 25 NSRs, 120 endpoints, 20 NATs
case-E: 30 NSRs, 140 endpoints, 25 NATs

Validation – Optimality



(a) Results with ratio 2

(b) Results with ratio 10

- Ratio used for weights govern the balance between the two optimization objectives.
- With large value, number of used resources steeply increase to meet the goal of configuration state preservation.

Conclusions and Future Work

- A new approach has been proposed to optimize FW reconfiguration, combining **automation**, **formal verification** and **optimization**.
- Achieved better scalability with respect to the state-of-the-art in the reconfiguration scenario.
- We will try to use this work to design a parallelized approach for the resolution of the problem, dividing the network into independent areas that could be configured separately and in parallel.

Thanks for your attention!

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GitHub: